Question:
Assume that B Corporation has the following total profit function equation: Total Profit = \pi = - $3,000 - $2,400 Q + $350Q^2 - $8.333Q^3 a) Compute the marginal profit b) Determine its output quantities
Answer Choices:
A. 22 and 7 units
B. 21 and 8 units
C. 23 and 6 units
D. 26 and 2 units
E. 24 and 4 units
F. 25 and 3 units
G. 20 and 5 units
H. 27 and 1 unit
I. 18 and 10 units
J. 19 and 9 units

Reasoning + Answer:

<reasoning process here>
We are given the total profit function:

ÏĢ = -3000 -2400Q +350QÂ² -8.333QÂ³

a) Compute the marginal profit.

Marginal profit (MP) is the derivative of total profit with respect to Q:

MP = dÏĢ/dQ

Differentiate term by term:

- d/dQ(-3000) = 0
- d/dQ(-2400Q) = -2400
- d/dQ(350QÂ²) = 700Q
- d/dQ(-8.333QÂ³) = -25QÂ³? Wait âĢĶ actually:

d/dQ(-8.333QÂ³) = -8.333 ÃĹ3QÂ² = -25QÂ² (since 8.333 ÃĹ3 = 25)

So:

MP = -2400 +700Q -25QÂ²

Therefore, marginal profit function is:

MP = -2400 +700Q -25QÂ²

b) Determine its output quantities.

The question says "determine its output quantities" âĢĶ likely meaning the quantity(ies) where marginal profit is zero (since that's typical for profit maximization).

Set marginal profit equal to zero:

-2400 +700Q -25QÂ² =0

Multiply both sides by -1:

2400 -700Q +25QÂ² =0

Rewriting:

25QÂ² -700Q +2400 =0

Divide entire equation by 25 to simplify:

QÂ² -28Q +96 =0

Now solve the quadratic equation:

Q = [28 Â± âĪļ(28Â² -4ÃĹ1ÃĹ96)] / 2
= [28 Â± âĪļ(784 -384)] / 2
= [28 Â± âĪļ400] / 2
= [28 Â± 20] / 2

So:

Q = (28 +20)/2 =48/2 =24
Q = (28 -20)/2 =8/2 =4

So the two output quantities are Q =24 and Q =4

Therefore, the pair is 24 and 4 units.

Looking at the answer choices, that matches:

E. 24 and 4 units

Note: The question says "output quantities", plural âĢĶ and the two roots of the marginal profit function are at Q =24 and Q =4 âĢĶ likely meaning the profit-maximizing quantities.

Also, since marginal profit is a downward opening parabola, the maximum occurs at the vertex or where MP=0, and we have two roots âĢĶ so possibly two break-even or profit-extremum points âĢĶ but the typical interpretation for "determine its output quantities" in such problems is the values of Q where marginal profit is zero âĢĶ so we take both roots.

Thus, the correct answer is the pair (24,4)

</reasoning process here>
<answer>E</answer>